Danish Nadeem B.Eng.

Mechatronics Engineering Graduate

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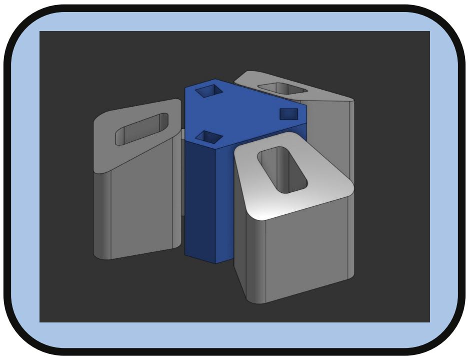
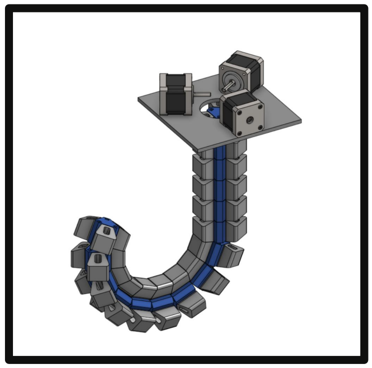
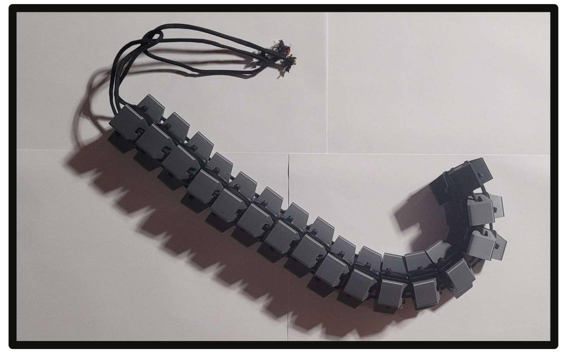
3D Octopus Arm End Effector







What?

- Redesigned an octopus-inspired robotic arm
- Cut manufacturing cost and assembly time

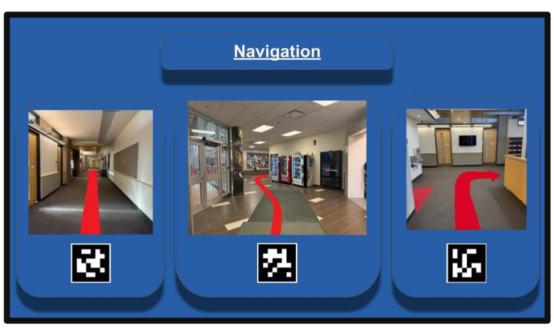
How?

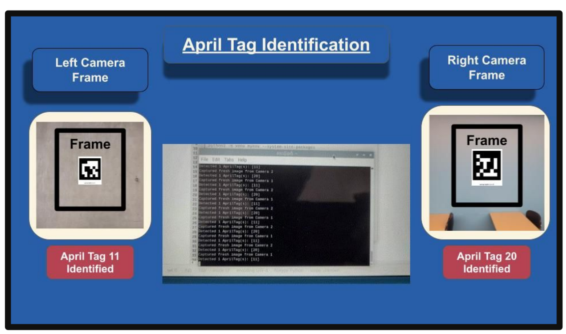
- Converted the TPU-based design to PLA,
- Simplifying the structure and reducing component count

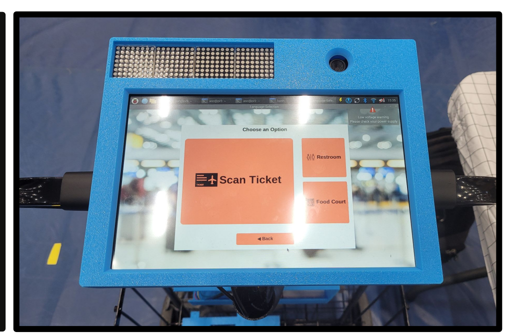
Resulted?

- Improved serviceability
- Reduced manufacturing cost
- Reduced assembly complexity
- Increasing compression strength, load capacity

Smart Airport Cart







What?

- Led a team of four to design and implement a marker-based indoor navigation system to guide elderly users through airport terminals.

How?

- Developed a touchscreen-based interface for user feedback and location instruction, focusing on clarity and accessibility.

Resulted?

- Presented to a public audiences during the capstone exhibition, highlighting strong communication and public-facing technical explanation.

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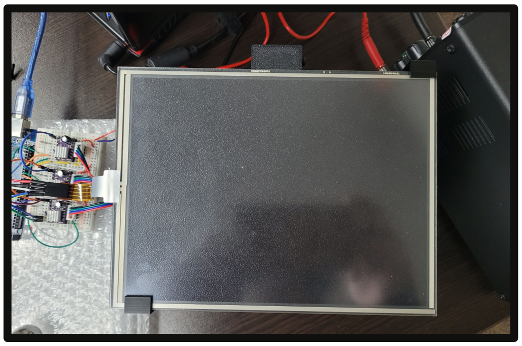
Ball Balancing using a 3RRS Parallel Manipulator



What?

- Optimized the program to reduce computational load, enabling the system to operate on an Arduino Mega (16 MHz) instead of a Teensy (600 MHz).

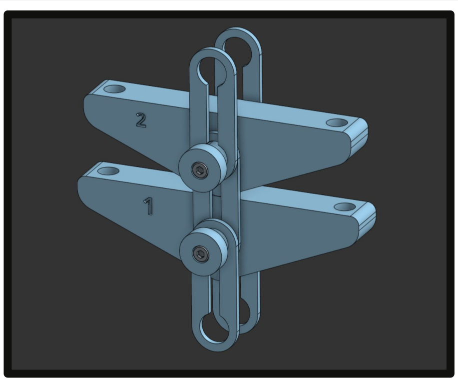
How?

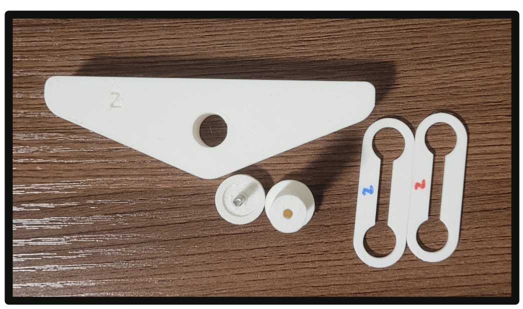
- Used direct port manipulation for the Arduino Mega to improve data acquisition speed, and applied signal filtering to eliminate false readings.

Resulted?

- Achieved project success by leading the team to have the manipulator balance the ball in a live demo

2D Octopus Arm End Effector





What?

- Redesigned the CAD model based on the research paper, and improved it by developing components suitable for manufacturing, assembly, and maintenance.

How?

- Removed print in place design from the CAD model in place for a composite assembly. Iterated the design through testing the CAD assembly, and 3D-printed models.

Resulted?

- Improved serviceability
- Reduced manufacturing cost
- Reduced assembly complexity
- Increasing compression strength, load capacity